

Virtualization -- Guided Notes

Unit tp-2.10 | CompTIA Network+ N10-009 Obj. FC0-U71 2.10 | N10-008 FC0-U71 2.10

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Virtualization allows one physical machine (the _____) to run multiple simulated environments called _____ (VMs). A _____ is the software layer that creates and manages VMs. Examples include VMware vSphere, Microsoft Hyper-V, and _____.

Question 2

A Type 1 hypervisor runs _____ on bare metal hardware, without a host OS underneath. Examples include VMware _____ and Microsoft _____. A Type 2 hypervisor runs on top of an existing _____ and is used on workstations. Examples include _____ and VirtualBox.

Question 3

Server consolidation uses virtualization to run many VMs on fewer _____ machines. If a company runs 20 servers at 5% CPU utilization each, they waste _____ capacity. Consolidating them onto fewer powerful physical hosts improves _____ utilization and reduces hardware, power, and _____ costs.

Question 4

A VM _____ is a saved state of a VM at a point in time. If a software update or configuration change breaks the VM, the administrator can _____ to the snapshot and restore the prior state. This is a powerful feature for _____ environments where changes carry risk.

Question 5

Containers (like _____) differ from VMs because they share the _____ kernel of the host OS rather than running a separate OS per instance. Containers are _____ (more/less) lightweight and start _____ (faster/slower) than VMs. They are commonly used in microservices and cloud deployments.

Question 6

VDI (Virtual Desktop Infrastructure) hosts users _____ on a central server. End users connect with _____ clients or thin clients. The advantage is that data and processing stay in the _____, making it secure and easy to manage. The risk is that if the server goes down, _____ users lose access.

Question 7

VM isolation means that a _____, crash, or attack inside one VM does not _____ other VMs running on the same host (under normal conditions). Each VM has _____ virtual hardware resources. This containment is what makes virtualization valuable for running _____ or testing malware.

Question 8 -- Real World Application

REAL WORLD: A cybersecurity teacher wants students to practice using Kali Linux for penetration testing exercises. The school computers run Windows 10 and cannot be reimaged. The teacher does not want students reachable from the school network during labs. Explain how virtualization solves all three constraints and describe the specific configuration you would set up.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-2-10-virtualization/tp-2-10-virtualization-answer-key.pdf
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